Kibana's Discover tab showing a KQL query

Elastic introduced KQL for querying logs and metrics in Elasticsearch indexes through Kibana. KQL is a flexible and intuitive language supporting a wide range of search capabilities, including field-level searches, logical operations, and complex queries. We can search using wildcards and, as shown in the example query, combine multiple search queries with logical operators such as `AND`, `OR`, and `NOT`. KQL also provides auto-completion and syntax highlighting, making it easier to construct and understand queries. The principle behind KQL uses the URI request API. It is handy to use URI-type queries without having to worry about constructing a query in Query DSL mode—if only there were an approach that gave us the best of both worlds.

The good news is that we can achieve the same URI functionality using a special query called `query_string`. This query type lets us define a query using logical operators in a request body.

10.9 The query_string query

The `query_string` type lets us construct a query using operators such as `AND` or `OR`, as well as logical operators such as `>` (greater than), `<=` (less than or equal to), `*` (contains in), and so on. This is easily understood with an example, so let's jump right into the code.

In the sidebar in the previous section, we fetched Bert's books using a long query on Kibana's Discover tab: `title:Java and author :Bert and edition:2 and release_date>=2000-01-01`. We can achieve this by writing a `query_string` query.

Listing 10.20 Creating a query string with operators

```
GET books/_search
{
  "query": {
    "query_string": {
      "query": "author:Bert AND edition:2 AND release_date>=2000-01-01"
    }
  }
}
```

Specifies the query_string query

Provides the search criteria in the query

The `query_string` query expects a `query` parameter where we provide the criteria. The query is constructed as name-value pairs: in this example, `author` is the field, and `Bert` is the value. Looking at the code, notice the following:

- The search query is built using Query DSL syntax (the `GET` request has a body).
- The search criteria are written to concatenate the fields using operators.

The query is as simple as writing a question in plain English. We can create complex criteria using these operators (in plain English) and provide them to the engine to get results for us.